合肥大學
HEFEI UNIVERSITY

Programming with Python

23. Schleifen mit `for`

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Programming with Python



Dies ist ein Kurs über das Programmieren mit der Programmiersprache Python an der Universität Hefei (合肥大学).

Die Webseite mit dem Lehrmaterial dieses Kurses ist https://thomasweise.github.io/programmingWithPython (siehe auch den QR-Kode unten rechts). Dort können Sie das Kursbuch (in Englisch) und diese Slides finden. Das Repository mit den Beispielprogrammen in Python finden Sie unter https://github.com/thomasWeise/programmingWithPythonCode.



Outline



1. Einleitung
2. Die `for`-Schleife
3. `continue` und `break`
4. Schleifen verschachteln
5. Iterieren über Sequenzen
6. Zusammenfassung



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- Danach sind wir im Prinzip in der Lage, *jede* Berechnung durchzuführen, die mit unseren Computern möglich ist.
- Alles was dann folgt sind nur Konzepte, die dafür da sind, unser Leben als Programmierer leichter zu machen.

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- Nach dem Schleifenkörper folgt eine Leerzeile, nach der es dann mit normalem Kode weitergeht.

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- Ranges, wie Slices, können auch mit negativen Schritten arbeiten: `range(40, 30, -3)` beginnt mit 40, steigt in Schritten von 3 *ab*, und hört *vor* 30 auf, ist also $(40, 37, 34, 31)$.

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"""Apply a for loop over a range."""

# We will construct a dictionary holding square numbers.
squares: dict[int, int] = {}  # Initialize `squares` as empty dict.

for i in range(5):  # i takes on the values 0, 1, 2, 3, 4 -- one by one.
    squares[i] = i * i  # Stores 0: 0, 1: 1, 2: 4, 3: 9, 4: 16.

for i in range(6, 9):  # i takes on the values 6, 7, and 8 one by one.
    squares[i] = i * i  # Stores 6: 36, 7: 49, 8: 64.

for i in range(20, 27, 2):  # i takes on the values 20, 22, 24, and 26.
    squares[i] = i * i      # Stores 20: 400, 22: 484, 24: 576, 26: 676.

for i in range(40, 30, -3):  # i takes on the values 40, 37, 34, and 31.
    squares[i] = i * i  # Stores 40: 1600, 37: 1369, 34: 1156, 31: 961.

print(squares)  # Print the dictionary.

for _ in range(3):  # Iterate the loop three times. Ignore counter `_`.
    print("Hello World!")  # We don't need the counter.
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↓ python3 for_loop_range.py ↓

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- In der ersten Schleife iteriert `i` über `range(5)`.

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for i in range(40, 30, -3):  # i takes on the values 40, 37, 34, and 31.
    squares[i] = i * i  # Stores 40: 1600, 37: 1369, 34: 1156, 31: 961.

print(squares)  # Print the dictionary.

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↓ `python3 for_loop_range.py` ↓

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Hello World!
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Beispiel (1)

- Zuerst wollen wir ein Dictionary bauen, in dem einige Ganzzahlen ihrem jeweiligen Quadrat zugeordnet sind.
- Wir benutzen vier `for`-Schleifen, um dieses Dictionary mit Daten zu füllen.
- Wir benutzen jeweils `i` als Schleifenvariable.
- In der ersten Schleife iteriert `i` über `range(5)`.
- Bei der ersten Ausführung des Schleifenkörpers hat `i` den Wert `0`.

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"""Apply a for loop over a range."""

# We will construct a dictionary holding square numbers.
squares: dict[int, int] = {}  # Initialize `squares` as empty dict.

for i in range(5):  # i takes on the values 0, 1, 2, 3, 4 -- one by one.
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for i in range(6, 9):  # i takes on the values 6, 7, and 8 one by one.
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- Wir benutzen jeweils `i` als Schleifenvariable.
- In der ersten Schleife iteriert `i` über `range(5)`.
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- Der Schleifenkörper `squares[i] = i * i` ist also äquivalent zu `squares[0] = 0` und weist daher den Wert `0` dem Schlüssel `0` im Dictionary `squares` zu.

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- In der dritten Iteration hat `i` den Wert `2` und wir führen `squares[2] = 4` aus.

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- Danach gilt `i = 3` und wir machen `squares[3] = 9`.

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- Zuletzt erfolgt dann `squares[4] = 16` und die erste Schleife ist fertig.

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- `i` wird daher zuerst `40`, dann `37`, dann `34`, und schließlich `31`.
- Nun drucken wir das Dictionary aus und bekommen das erwartete Ergebnis.

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- Was wir in diesem Programm wieder und wieder tun, ist...

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from math import pi, sqrt   # We need sqrt. pi is for comparison.

# We use f-strings with Unicode escapes to print the current result.
# "\u03c0" is the Unicode escape for the Greek letter pi.
# "\u2248" is the Unicode escape for the "approximately equal" sign.
print(f"We use Liu Hui's Method to Approximate \u03c0\u2248{pi}.")
e = 6     # the number of edges: We start with a hexagon, i.e., e=6.
s = 1.0  # the side length: Initially 1, meaning the radius is also 1.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2   # We double the number of edges... ...now there are 12.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 24.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 48.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 96.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 192.
s = sqrt(2 - sqrt(4 - (s ** 2)))
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")
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↓ python3 pi_liu_hui.py ↓

```
We use Liu Hui's Method to Approximate π≈3.141592653589793.
6 edges, side length=1.0 give π≈3.0.
12 edges, side length=0.5176380902050416 give π≈3.1058285412302498.
24 edges, side length=0.2610523844401031 give π≈3.132628613281237.
48 edges, side length=0.13080625846028635 give π≈3.139350203046872.
96 edges, side length=0.0654381656435527 give π≈3.14103195089053.
192 edges, side length=0.03272346325297234 give π≈3.1414524722853443.
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- Benutzen wir nun diese neuen Informationen, um unser Programm zum Annähern von π zu verbessern.
- Was wir in diesem Programm wieder und wieder tun, ist:
  - Wir verdoppeln die Anzahl `e` der Seiten eines regelmäsigen `e`-gons mit `e *= 2`.

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s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 192.
s = sqrt(2 - sqrt(4 - (s ** 2)))
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")
```

↓ `python3 pi_liu_hui.py` ↓

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- Was wir in diesem Programm wieder und wieder tun, ist:
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from math import pi, sqrt   # We need sqrt. pi is for comparison.

# We use f-strings with Unicode escapes to print the current result.
# "\u03c0" is the Unicode escape for the Greek letter pi.
# "\u2248" is the Unicode escape for the "approximately equal" sign.
print(f"We use Liu Hui's Method to Approximate \u03c0\u2248{pi}.")
e = 6     # the number of edges: We start with a hexagon, i.e., e=6.
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print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2   # We double the number of edges... ...now there are 12.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 24.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

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e *= 2  # We double the number of edges... ...now there are 192.
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e *= 2   # We double the number of edges... ...now there are 12.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")

e *= 2  # We double the number of edges... ...now there are 24.
s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length.
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- Wenn wir all das einfach in eine Schleife packen...

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"""We execute Liu Hui's method to approximate pi in a loop."""
from math import pi, sqrt

print(f"We use Liu Hui's Method to Approximate \u03c0\u2248{pi}.")
e: int = 6  # the number of edges: We start with a hexagon, i.e., e=6.
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for _ in range(6):
    print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")
    e *= 2  # We double the number of edges...
    s = sqrt(2 - sqrt(4 - (s ** 2)))  # ...and recompute the side length
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- Und die Ausgabe ist die gleiche!

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continue und break



`continue` und `break`

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`continue` **und** `break`



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- Es ist nicht ungewöhnlich dass wir manchmal nach ein paar Berechnungen im Körper einer Schleife schon wissen, dass wir mit der nächsten Iteration weitermachen können, anstatt den Rest des Schleifenkörpers auszuführen.

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- Für den ersten Fall gibt es das Schlüsselwort `continue`, für den zweiten Fall das Schlüsselwort `break`[56].

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for i in range(15):  # i takes on the values from 0 to 14 one by one.
    s: str = f"i is now {i}."  # Create a string with the value of i.
    if i > 10:  # If i is greater than 10, then...
        break  # ...abort the loop altogether, do not execute next line.
    if 5 <= i <= 8:  # If i is in the range of 5..8, then...
        continue  # ...skip the rest of the loop body, do next iteration
    print(s)  # We get here if neither continue nor break were called.

print("All done.")  # We always get here.
```

↓ `python3 for_loop_continue_break.py` ↓

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- Es geht dann nach der Schleife weiter, also mit `print("All done.")`.

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All done.
```

Beispiel

- Also wenn `i > 10`, dann wird `break` ausgeführt, was die Schleife dann sofort beendet.
- Die aktuelle Iteration wird dann nicht mehr beendet und der Kode im restlichen Schleifenkörper wird überprungen und die Schleife verlassen.
- Es geht dann nach der Schleife weiter, also mit `print("All done.")`.
- Wenn `i > 10` aber nicht zutrifft, dann wird das `break` nicht ausgeführt aber stattdessen der restliche Kode im Schleifenkörper.

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"""Here we explore the `break` and `continue` statements."""

for i in range(15):  # i takes on the values from 0 to 14 one by one.
    s: str = f"i is now {i}."  # Create a string with the value of i.
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- Natürlich wird die Schleife abbrechen, sobald `i == 11`.

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- Das Programm wird also `s` nur für $i \in 0..4 \cup \{9, 10\}$ ausgeben, bevor letztendlich `All done.` ausgegeben wird.

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- Das Programm wird also `s` nur für $i \in 0..4 \cup \{9, 10\}$ ausgeben, bevor letztendlich `All done.` ausgegeben wird.
- Mit `break` und `continue` haben wir nun zwei neue Werkzeuge, mit denen wir entweder die die ganze Schleife abbrechen können oder die aktuelle Iteration beenden (und mit der nächsten weitermachen können, wenn es eine gibt).

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Definition: Primezahl

Eine Primzahl (EN: *prime number*) $p \in \mathbb{N}_1$ ist eine positive Ganzzahl $p > 1$, also größer als eins, die keine positiven ganzzahligen Teiler anders als 1 und p selbst hat[20,69,92].

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Beispiel

- In unserem Programm wollen wir alle Primzahlen $p < 200$ in einer Liste `primes` speichern.

```python
"""Compute all primes less than 200 using two nested for loops."""

from math import isqrt  # the integer square root == int(sqrt(...))

primes: list[int] = [2]  # the list for the primes; We know 2 is prime.
n_divisions: int = 0  # We want to know how many divisions we needed.

for candidate in range(3, 200, 2):  # ...all odd numbers less than 200.
    is_prime: bool = True  # Let us assume that `candidate` is prime.

    for check in range(3, isqrt(candidate) + 1, 2):  # ...odd numbers
        n_divisions += 1  # Every test requires one modulo division.
        if candidate % check == 0:  # modulo == 0: division without rest
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# Finally, print the list of prime numbers.
print(f"After {n_divisions} divisions: {len(primes)} primes {primes}.")
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↓ python3 for_loop_nested_primes.py ↓

```
After 252 divisions: 46 primes [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
↪ 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103,
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- Wir setzen also anfangs `primes = [2]`.
- Dann brauchen wir uns nachher nur noch um ungerade Zahlen aus dem Intervall 3..199 zu kümmern.

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Beispiel

- Wir wissen, dass `2` eine Primzahl ist – und zwar die einzige gerade Primzahl – also können wir sie direkt bei der Initialisierung der Liste hineinschreiben.
- Wir setzen also anfangs `primes = [2]`.
- Dann brauchen wir uns nachher nur noch um ungerade Zahlen aus dem Intervall 3..199 zu kümmern.
- Aus Interesse wollen wir auch die Anhahl der Divisionen in der Variable `n_divisions` mitzählen, die wir machen, bis wir die ganze Liste von Primzahlen zusammenhaben.

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"""Compute all primes less than 200 using two nested for loops."""

from math import isqrt  # the integer square root == int(sqrt(...))

primes: list[int] = [2]  # the list for the primes; We know 2 is prime.
n_divisions: int = 0  # We want to know how many divisions we needed.

for candidate in range(3, 200, 2):  # ...all odd numbers less than 200.
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- Natürlich müssen diese auch größer oder gleich 3 sein.

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Beispiel

- Wir werden diesen Plan mit einer geschachtelten / inneren Schleife implementieren.
- Weil die Schleifenvariable `candidate` immer ungerade ist, kommen auch nur ungerade Zahlen als Teiler in Frage.
- Natürlich müssen diese auch größer oder gleich 3 sein.
- Wir müssen auch nur dann prüfen, ob eine Zahl `check` ein Teiler von `candidate`, wenn sie nicht größer als $\sqrt{\texttt{candidate}}$ ist.

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After 252 divisions: 46 primes [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
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Beispiel

- Darum können wir unsere zweite, innere Schleife mit der Schleifenvariablen `check` über `range(3, isqrt(candidate)+ 1, 2)` iterieren lassen.
- Für alle `candidate <= 8` wird diese Schleife gar nicht ausgeführt, denn `isqrt(8)== 2`.
- Dann bleibt `is_prime` `True` und wir hängen `candidate` in `primes` an.
- Für `candidate > 8`, geht `check` dann von 3 bis $\lfloor\sqrt{\texttt{candidate}}\rfloor$.

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"""Compute all primes less than 200 using two nested for loops."""

from math import isqrt  # the integer square root == int(sqrt(...))

primes: list[int] = [2]  # the list for the primes; We know 2 is prime.
n_divisions: int = 0  # We want to know how many divisions we needed.

for candidate in range(3, 200, 2):  # ...all odd numbers less than 200.
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    for check in range(3, isqrt(candidate) + 1, 2):  # ...odd numbers
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- Wir erfahren, dass wir mit 252 Divisionen alle 46 Primzahlen im Intervall 2..199 finden konnten.

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            break  # candidate is not prime. We can stop the inner loop.

    if is_prime:  # If True: no smaller number divides candidate evenly.
        primes.append(candidate)  # Store candidate in primes list.

# Finally, print the list of prime numbers.
print(f"After {n_divisions} divisions: {len(primes)} primes {primes}.")
```

↓ python3 for_loop_nested_primes.py ↓

```
After 252 divisions: 46 primes [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
↪ 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103,
↪ 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173,
↪ 179, 181, 191, 193, 197, 199].
```

Beispiel

- Wenn ja, dann hängen wir `candidate` an die Liste `primes` an, in dem wir die `append`-Methode der Liste aufrufen.
- Nach dem wir mit der äußeren Schleife fertig sind, drucken wir sowohl die Anzahl `n_divisions` der benötigten Divsionen, die Anzahl `len(primes)` der gefundenen Primzahlen, als auch die Liste `primes` der Primzahlen selbst aus.
- Wir erfahren, dass wir mit 252 Divisionen alle 46 Primzahlen im Intervall 2..199 finden konnten.
- (OK, wir haben ignoriert bzw. wissen nicht, ob `isqrt` irgendwelche Divisionen durchführt, aber egal.).

```python
"""Compute all primes less than 200 using two nested for loops."""

from math import isqrt  # the integer square root == int(sqrt(...))

primes: list[int] = [2]  # the list for the primes; We know 2 is prime.
n_divisions: int = 0  # We want to know how many divisions we needed.

for candidate in range(3, 200, 2):  # ...all odd numbers less than 200.
    is_prime: bool = True  # Let us assume that `candidate` is prime.

    for check in range(3, isqrt(candidate) + 1, 2):  # ...odd numbers
        n_divisions += 1  # Every test requires one modulo division.
        if candidate % check == 0:  # modulo == 0: division without rest
            is_prime = False  # check divides candidate evenly, so
            break  # candidate is not prime. We can stop the inner loop.

    if is_prime:  # If True: no smaller number divides candidate evenly.
        primes.append(candidate)  # Store candidate in primes list.

# Finally, print the list of prime numbers.
print(f"After {n_divisions} divisions: {len(primes)} primes {primes}.")
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Iterieren über Sequenzen



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- Wir haben auch gesagt, dass wir schon verschiedene Kontainerdatentypen kennen, die wir als Sequenzen behandeln können.
- Wir müssten also über `lists`, `tuples`, `sets`, und `dicts` iterieren können...

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- In unserem Beispielprogramm bauen wir uns eine Liste `txt` mit Strings zusammen, die wir später ausgeben wollen.

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"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
    txt.append(f"{i = }")  # We store "i = 1", "i = 2", "i = 3"...

tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
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st: set[str] = {"u", "v", "w"}  # Create a set with 3 strings.
for s in st:  # s takes on the values "u", "v", and "w" (unordered!).
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dc: dict[float, bool] = {1.1: True, 2.5: False}  # Create a dictionary.
for k in dc:  # Iterate over the keys in the dictionary == 1.1 and 2.5.
    txt.append(f"{k = }")  # We store "k=1.1" and "k=2.5".
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# Merge text into single string with separator ", " and print it.
print(", ".join(txt))
```

↓ `python3 for_loop_sequence.py` ↓

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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
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- In unserem Beispielprogramm bauen wir uns eine Liste `txt` mit Strings zusammen, die wir später ausgeben wollen.
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Beispiel (1)



- Diese Strings werden alle via `append` an die Liste `txt` angehängt.
- Machen wir nun mit einem Tupel `tp` mit den viel Fließkommazahlen `(7.6, 9.4, 8.1)` weiter.
- Hier funktioniert es ganz genauso.
- `for f in tp` lässt die Schleifenvariable `f` nacheinander die Werte `7.6`, `9.4`, und `8.1` annehmen.

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↓ `python3 for_loop_sequence.py` ↓

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- Der `!r` Format-Specifier convertiert die Werte von `s` zu ihrer *Repräsentation*, was im Grunde Anführungszeichen um die Strings setzt (und ggf. Escape-Sequenzen für nicht-druckbare Zeichen benutzt).

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↓ `python3 for_loop_sequence.py` ↓

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Beispiel (1)



- Wieder speichern wir diese als schön formatierte Texte in der Liste `txt`.
- Diesmal benutzen wir den f-String `f"s = {s!r}"`.
- Der `!r` Format-Specifier convertiert die Werte von `s` zu ihrer *Repräsentation*, was im Grunde Anführungszeichen um die Strings setzt (und ggf. Escape-Sequenzen für nicht-druckbare Zeichen benutzt).
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for k in dc:  # Iterate over the keys in the dictionary == 1.1 and 2.5.
    txt.append(f"{k = }")  # We store "k=1.1" and "k=2.5".
for v in dc.values():       # Iterate over the values in the dictionary.
    txt.append(f"{v = }")  # We store "v=True" and "v=False".
for k, v in dc.items():     # Iterate over the key-value combinations.
    txt.append(f"{k}: {v}")  # Store "1.1: True" and "2.5: False"

# Merge text into single string with separator ", " and print it.
print(", ".join(txt))
```

↓ `python3 for_loop_sequence.py` ↓

```
i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
    ↪ False
```

Beispiel (1)



- Das ist das selbe, als über `dc.keys()` zu iterieren.
- Iterieren wir über `dc.values()`, dann sehen wir alle Werte in `dc`.
- Iterieren wir über `dc.items()`, dann sehen wir alle Schlüssel-Wert-Paare als Tupels.
- Wir probieren alle drei Varianten aus.

```python
"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
    txt.append(f"{i = }")  # We store "i = 1", "i = 2", "i = 3"...

tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
    txt.append(f"{f = }")  # We store "f = 7.6", "f = 9.4", ...

st: set[str] = {"u", "v", "w"}  # Create a set with 3 strings.
for s in st:  # s takes on the values "u", "v", and "w" (unordered!).
    txt.append(f"s = {s!r}")  # We store "s = 'u'", "s = 'v'", ...

dc: dict[float, bool] = {1.1: True, 2.5: False}  # Create a dictionary.
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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
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Beispiel (1)



- Iterieren wir über `dc.values()`, dann sehen wir alle Werte in `dc`.
- Iterieren wir über `dc.items()`, dann sehen wir alle Schlüssel-Wert-Paare als Tupels.
- Wir probieren alle drei Varianten aus.
- Wir iterieren zuerst über die Schlüssel mit `for k in dc`, wodurch `k` erst den Wert `1.1` und dann `2.5` annimmt.

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"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
    txt.append(f"{i = }")  # We store "i = 1", "i = 2", "i = 3"...

tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
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Beispiel (1)



- Iterieren wir über `dc.items()`, dann sehen wir alle Schlüssel-Wert-Paare als Tupels.
- Wir probieren alle drei Varianten aus.
- Wir iterieren zuerst über die Schlüssel mit `for k in dc`, wodurch `k` erst den Wert `1.1` und dann `2.5` annimmt.
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"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
    txt.append(f"{i = }")  # We store "i = 1", "i = 2", "i = 3"...

tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
    ↪ False
```

Beispiel (1)

- Wir probieren alle drei Varianten aus.
- Wir iterieren zuerst über die Schlüssel mit `for k in dc`, wodurch `k` erst den Wert `1.1` und dann `2.5` annimmt.
- Wir iterieren über die Werte mit `for v in dc.values()`, wodurch `v` erst `True` und dann `False` wird.
- Zu guter Letzt iterieren wir über die Schlüssel-Wert-Paare.

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"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
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tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
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st: set[str] = {"u", "v", "w"}  # Create a set with 3 strings.
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    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
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Beispiel (1)



- Wir iterieren zuerst über die Schlüssel mit `for k in dc`, wodurch `k` erst den Wert `1.1` und dann `2.5` annimmt.
- Wir iterieren über die Werte mit `for v in dc.values()`, wodurch `v` erst `True` und dann `False` wird.
- Zu guter Letzt iterieren wir über die Schlüssel-Wert-Paare.
- Schauen Sie genau hin!

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Beispiel (1)



- Wir iterieren über die Werte mit `for v in dc.values()`, wodurch `v` erst `True` und dann `False` wird.
- Zu guter Letzt iterieren wir über die Schlüssel-Wert-Paare.
- Schauen Sie genau hin!
- Wir *könnten* schreiben `for t in dc.items()`, wodurch wir eine Variable `t` die Werte `(1.1, True)` und dann `2.5: False` annehmen würde.

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    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
    ↪  False
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Beispiel (1)



- Zu guter Letzt iterieren wir über die Schlüssel-Wert-Paare.
- Schauen Sie genau hin!
- Wir *könnten* schreiben `for t in dc.items()`, wodurch wir eine Variable `t` die Werte `(1.1, True)` und dann `2.5: False` annehmen würde.
- Aber wir haben ja vom automatischen „Auspacken“ von Tupeln gelernt.

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Beispiel (1)



- Schauen Sie genau hin!
- Wir *könnten* schreiben `for t in dc.items()`, wodurch wir eine Variable `t` die Werte `(1.1, True)` und dann `2.5: False` annehmen würde.
- Aber wir haben ja vom automatischen „Auspacken“ von Tupeln gelernt.
- Stattdessen schreiben wir also `for k, v in dc.items()` hin.

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Beispiel (1)



- Wir *könnten* schreiben `for t in dc.items()`, wodurch wir eine Variable `t` die Werte `(1.1, True)` und dann `2.5: False` annehmen würde.
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Beispiel (1)



- Aber wir haben ja vom automatischen „Auspacken“ von Tupeln gelernt.
- Stattdessen schreiben wir also `for k, v in dc.items()` hin.
- Das ist eine sowas wie eine Akürzung für `for t in dc.items()` gefolgt von `k, v = t`.
- Es packt die Tupels der in der Sequenz `dc.items` direkt aus.

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↓ `python3 for_loop_sequence.py` ↓

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Beispiel (1)



- Stattdessen schreiben wir also `for k, v in dc.items()` hin.
- Das ist eine sowas wie eine Akürzung für `for t in dc.items()` gefolgt von `k, v = t`.
- Es packt die Tupels der in der Sequenz `dc.items` direkt aus.
- Wir bekommen also Paare `k=1.1`, `v=True` und `k=2.5`, `v=False`.

```python
"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
    txt.append(f"{i = }")  # We store "i = 1", "i = 2", "i = 3"...

tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
    txt.append(f"{f = }")  # We store "f = 7.6", "f = 9.4", ...

st: set[str] = {"u", "v", "w"}  # Create a set with 3 strings.
for s in st:  # s takes on the values "u", "v", and "w" (unordered!).
    txt.append(f"s = {s!r}")  # We store "s = 'u'", "s = 'v'", ...

dc: dict[float, bool] = {1.1: True, 2.5: False}  # Create a dictionary.
for k in dc:  # Iterate over the keys in the dictionary == 1.1 and 2.5.
    txt.append(f"{k = }")  # We store "k=1.1" and "k=2.5".
for v in dc.values():      # Iterate over the values in the dictionary.
    txt.append(f"{v = }")  # We store "v=True" and "v=False".
for k, v in dc.items():    # Iterate over the key-value combinations.
    txt.append(f"{k}: {v}")  # Store "1.1: True" and "2.5: False"

# Merge text into single string with separator ", " and print it.
print(", ".join(txt))
```

↓ `python3 for_loop_sequence.py` ↓

```
i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
    ↪ False
```

Beispiel (1)



- Das ist eine sowas wie eine Akürzung für `for t in dc.items()` gefolgt von `k, v = t`.
- Es packt die Tupels der in der Sequenz `dc.items` direkt aus.
- Wir bekommen also Paare `k=1.1`, `v=True` und `k=2.5`, `v=False`.
- Und wieder hängen wir diese als Text an unsere Liste `txt` an.

```python
"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
    txt.append(f"{i = }")  # We store "i = 1", "i = 2", "i = 3"...

tp: tuple[float, ...] = (7.6, 9.4, 8.1)  # Create a tuple with 3 floats.
for f in tp:  # i takes on the values 7.6, 9.4, and 8.1.
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# Merge text into single string with separator ", " and print it.
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↓ `python3 for_loop_sequence.py` ↓

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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
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Beispiel (1)



- Es packt die Tupels der in der Sequenz `dc.items` direkt aus.
- Wir bekommen also Paare `k=1.1`, `v=True` und `k=2.5`, `v=False`.
- Und wieder hängen wir diese als Text an unsere Liste `txt` an.
- Nach diesen ganzen Schleifen haben wir nun eine Liste `txt` mit 16 Strings.

```python
"""Iterate over several different containers with `for` loops."""

txt: list[str] = []  # We will collect the output text in this list.

lst: list[int] = [1, 2, 3, 50]  # Create a list with 4 integers.
for i in lst:  # i takes on the values 1, 2, 3, and 50.
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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
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Beispiel (1)



- Wir bekommen also Paare `k=1.1`, `v=True` und `k=2.5`, `v=False`.
- Und wieder hängen wir diese als Text an unsere Liste `txt` an.
- Nach diesen ganzen Schleifen haben wir nun eine Liste `txt` mit 16 Strings.
- Wir wollen diese zu einzigen Zeichenkette zusammenfühgen, wobei wir `", "` als Separator benutzen wollen.

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Beispiel (1)



- Und wieder hängen wir diese als Text an unsere Liste `txt` an.
- Nach diesen ganzen Schleifen haben wir nun eine Liste `txt` mit 16 Strings.
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- Das könnten wir mit einer Schleife machen.

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Beispiel (1)



- Nach diesen ganzen Schleifen haben wir nun eine Liste `txt` mit 16 Strings.
- Wir wollen diese zu einzigen Zeichenkette zusammenfügen, wobei wir `", "` als Separator benutzen wollen.
- Das könnten wir mit einer Schleife machen.
- Python bietet aber eine einfachere und schnellere Methode dafür an.

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Beispiel (1)



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- Das könnten wir mit einer Schleife machen.
- Python bietet aber eine einfachere und schnellere Methode dafür an.
- Die Methode `join` der Klasse `str`.

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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
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Beispiel (1)



- Das könnten wir mit einer Schleife machen.
- Python bietet aber eine einfachere und schnellere Methode dafür an.
- Die Methode `join` der Klasse `str`.
- Für jeden String `z`, akzeptiert `z.join(seq)` eine Sequenz `seq` von Strings.

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txt: list[str] = []  # We will collect the output text in this list.

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# Merge text into single string with separator ", " and print it.
print(", ".join(txt))
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↓ `python3 for_loop_sequence.py` ↓

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i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
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```

Beispiel (1)



- Python bietet aber eine einfachere und schnellere Methode dafür an.
- Die Methode `join` der Klasse `str`.
- Für jeden String `z`, akzeptiert `z.join(seq)` eine Sequenz `seq` von Strings.
- Es hängt alle Strings in `seq` hintereinander an, wobei jeweils `z` als Separator zwischen zwei Strings eingefügt wird.

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Beispiel (1)



- Die Methode `join` der Klasse `str`.
- Für jeden String `z`, akzeptiert `z.join(seq)` eine Sequenz `seq` von Strings.
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- Daher produziert `", ".join(txt)` als Ergebnis `i = 1, i = 2, i = 3, i = 50, f = 7.6, ....`

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- Daher produziert `", ".join(txt)` als Ergebnis `i = 1, i = 2, i = 3, i = 50, f = 7.6, ...`.
- Dieser Text wird mit der `print` ausgegeben.

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    txt.append(f"s = {s!r}")  # We store "s = 'u'", "s = 'v'", ...

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for k in dc:  # Iterate over the keys in the dictionary == 1.1 and 2.5.
    txt.append(f"{k = }")  # We store "k=1.1" and "k=2.5".
for v in dc.values():      # Iterate over the values in the dictionary.
    txt.append(f"{v = }")  # We store "v=True" and "v=False".
for k, v in dc.items():    # Iterate over the key-value combinations.
    txt.append(f"{k}: {v}")  # Store "1.1: True" and "2.5: False"

# Merge text into single string with separator ", " and print it.
print(", ".join(txt))
```

↓ `python3 for_loop_sequence.py` ↓

```
i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'v', s = 'w
    ↪ ', s = 'u', k = 1.1, k = 2.5, v = True, v = False, 1.1: True, 2.5:
    ↪ False
```

Beispiel (2)

- Verwenden wir das Konzept der Iteration über Sequenzen nun für etwas Sinnvolles.

```python
"""Compute all primes less than 200 using two nested for loops."""

from math import isqrt  # the integer square root == int(sqrt(...))

primes: list[int] = [2]  # the list for the primes; We know 2 is prime.
n_divisions: int = 0  # We want to know how many divisions we needed.

for candidate in range(3, 200, 2):  # ...all odd numbers less than 200.
    is_prime: bool = True  # Let us assume that `candidate` is prime.

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Beispiel (2)

- `primes` beinhaltet ja alle Primzahlen, die kleiner als `candidate` sind.
- In einem effizienteren Programm ersetzen wir also einfach den Kopf der inneren Schleife mit `for check in primes:`.
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- Allerdings brauchen wir nun nur 224 Divisionen anstatt von 252.

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- Und wir können Schleifen und Alternativen beliebig ineinander verschachteln.

谢谢您们！

Thank you!

Vielen Dank!



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Glossary (in English) I



Bash is a shell used under Ubuntu Linux, i.e., the program that „runs“ in the terminal and interprets your commands, allowing you to start and interact with other programs[12,57,98]. Learn more at https://www.gnu.org/software/bash.

BCE The time notation *before Common Era* is a non-religious but chronological equivalent alternative to the traditional *Before Christ (BC)* notation, which refers to the years *before* the birth of Jesus Christ[16]. The years BCE are counted down, i.e., the larger the year, the farther in the past. The year 1 BCE comes directly before the year 1 Common Era (CE)[71,99].

C is a programming language, which is very successful in system programming situations[27,64,88].

CE The time notation *Common Era* is a non-religious but chronological equivalent alternative to the traditional *Anno Domini (AD)* notation, which refers to the years *after* the birth of Jesus Christ[16]. The years CE are counted upwards, i.e., the smaller they are, the farther they are in the past. The year 1 CE comes directly after the year 1 before Common Era (BCE)[71,99].

client In a client-server architecture, the client is a device or process that requests a service from the server. It initiates the communication with the server, sends a request, and receives the response with the result of the request. Typical examples of clients are web browsers in the internet as well as clients for database management systems (DBMSes), such as psql.

client-server architecture is a system design where a central server receives requests from one or multiple clients[7,50,61,65,68]. These requests and responses are usually sent over network connections. A typical example of such a system is the World Wide Web (WWW), where web servers host websites and make them available to web browsers, the clients. Another typical example is the structure of database (DB) software, where a central server, the DBMS, offers access to the DB to the different clients. Here, the client can be some terminal software shipping with the DBMS, such as psql, or the different applications that access the DBs.

DB A *database* is an organized collection of structured information or data, typically stored electronically in a computer system. Databases are discussed in our book *Databases*[87].

DBMS A *database management system* is the software layer located between the user or application and the DB. The DBMS allows the user/application to create, read, write, update, delete, and otherwise manipulate the data in the DB[96].

denominator The number b of a fraction $\frac{a}{b} \in \mathbb{Q}$ is called the *denominator*.

Glossary (in English) II



escape sequence Escaping is the process of presenting „forbidden“ characters or symbols in a sequence of characters or symbols. In Python[89], string escapes allow us to include otherwise impossible characters, such as string delimiters, in a string. Each such character is represented by an *escape sequence*, which usually starts with the backslash character („\“)[29]. In Python strings, the escape sequence `\"`, for example, stands for `"`, the escape sequence `\\` stands for `\`, and the escape sequence `\n` stands for a newline or linebreak character. In Python f-strings, the escape sequence `{{` stands for a single curly brace `{`. In PostgreSQL[87], similar C-style escapes (starting with „\“) are supported[80].

f-string let you include the results of expressions in strings[13,35–37,54,74]. They can contain expressions (in curly braces) like `f"a{6-1}b"` that are then transformed to text via (string) interpolation, which turns the string to `"a5b"`. F-strings are delimited by `f"..."`.

irrational An *irrational* number z is a real number that cannot be expressed as fraction $\frac{p}{q}$ of two integers $p \in \mathbb{Z}$ and $q \in \mathbb{N}_1$[5,90]. In other words, it holds that $z \in \mathbb{R}$ and $z \notin \mathbb{Q}$. The set of irrational numbers could be denoted as $\mathbb{R} \setminus \mathbb{Q}$. Irrational numbers have decimal expansions that neither terminate nor become periodic. Examples of irrational numbers are $\sqrt{2}$, ϕ, π, and e.

IT information technology

LAMP Stack A system setup for web applications: Linux, Apache (a web server), MySQL, and the server-side scripting language PHP[14,42].

Linux is the leading open source operating system, i.e., a free alternative to Microsoft Windows[3,41,72,85,86]. We recommend using it for this course, for software development, and for research. Learn more at https://www.linux.org. Its variant Ubuntu is particularly easy to use and install.

MariaDB An open source relational database management system that has forked off from MySQL[1,2,4,28,52,66]. See https://mariadb.org for more information.

Microsoft Windows is a commercial proprietary operating system[11]. It is widely spread, but we recommend using a Linux variant such as Ubuntu for software development and for our course. Learn more at https://www.microsoft.com/windows.

modulo division is, in Python, done by the operator `%` that computes the remainder of a division. `15 % 6` gives us `3`.

Glossary (in English) III



MySQL An open source relational database management system[10,28,67,82,95]. MySQL is famous for its use in the LAMP Stack. See `https://www.mysql.com` for more information.

numerator The number a of a fraction $\frac{a}{b} \in \mathbb{Q}$ is called the *numerator*.

PostgreSQL An open source object-relational DBMS[33,60,63,82]. See `https://postgresql.org` for more information.

psql is the client program used to access the PostgreSQL DBMS server.

Python The Python programming language[43,49,51,89], i.e., what you will learn about in our book[89]. Learn more at `https://python.org`.

relational database A relational DB is a database that organizes data into rows (tuples, records) and columns (attributes), which collectively form tables (relations) where the data points are related to each other[19,38,40,75,81,87,94].

server In a client-server architecture, the server is a process that fulfills the requests of the clients. It usually waits for incoming communication carring the requests from the clients. For each request, it takes the necessary actions, performs the required computations, and then sends a response with the result of the request. Typical examples of servers are web servers[14] in the internet as well as DBMSes. It is also common to refer to the computer running the server processes as server as well, i.e., to call it the „server computer“[48].

SQL The *Structured Query Language* is basically a programming language for querying and manipulating relational databases[17,22,24,25,44,55,76–78,81]. It is understood by many DBMSes. You find the Structured Query Language (SQL) commands supported by PostgreSQL in the reference[76].

(string) interpolation In Python, string interpolation is the process where all the expressions in an f-string are evaluated and the final string is constructed. An example of string interpolation is turning `f"Rounded {1.234:.2f}"` to `"Rounded 1.23"`.

Glossary (in English) IV



terminal A terminal is a text-based window where you can enter commands and execute them[3,18]. Knowing what a terminal is and how to use it is very essential in any programming- or system administration-related task. If you want to open a terminal under Microsoft Windows, you can Druck auf ⊞+R, dann Schreiben von `cmd`, dann Druck auf ↵. Under Ubuntu Linux, Ctrl+Alt+T opens a terminal, which then runs a Bash shell inside.

transcendental A *transcendental* number is a number that is not the root of any integer polynomial, meaning that it is not an algebraic number of any degree[93]. Every real transcendental number must also be irrational, since a rational number is, by definition, an algebraic number of degree one[93]. Examples of transcendental numbers are π and e.

Ubuntu is a variant of the open source operating system Linux[18,42]. We recommend that you use this operating system to follow this class, for software development, and for research. Learn more at https://ubuntu.com. If you are in China, you can download it from https://mirrors.ustc.edu.cn/ubuntu-releases.

WWW World Wide Web[6,26]

ϕ The golden ratio (or golden section) ϕ is the irrational number $\frac{1+\sqrt{5}}{2}$. It is the ratio of a line segment cut into two pieces of different lengths such that the ratio of the whole segment to that of the longer segment is equal to the ratio of the longer segment to the shorter segment[15,30]. The golden ratio is approximately $\phi \approx 1.618\,033\,988\,749\,894\,848\,204\,586\,834$[73]. Represented as `float` in Python, its value is `1.618033988749895`.

π is the ratio of the circumference U of a circle and its diameter d, i.e., $\pi = U/d$. $\pi \in \mathbb{R}$ is an irrational and transcendental number[34,45,58], which is approximately $\pi \approx 3.141\,592\,653\,589\,793\,238\,462\,643$. An early method to approximate π was developed by the Chinese mathematician LIU Hui (刘徽) based on inscribing regular e-gons into a circle somewhere in the third century CE[59,97] in his commentary to the famous Chinese mathematics book *Jiu Zhang Suanshu* (九章算术)[21,23,46,59,79]. In Python, π is provided by the `math` module as variable `pi` with value `3.141592653589793`. In PostgreSQL, it is provided by the SQL function `pi()` with value `3.141592653589793`[53].

$i..j$ with $i, j \in \mathbb{Z}$ and $i \leq j$ is the set that contains all integer numbers in the inclusive range from i to j. It always holds that $i..j \subset \mathbb{Z}$. For example, 5..9 is equivalent to $\{5, 6, 7, 8, 9\}$

Glossary (in English) V

e is Euler's number[32], the base of the natural logarithm. $e \in \mathbb{R}$ is an irrational and transcendental number[34,45], which is approximately $e \approx 2.718\,281\,828\,459\,045\,235\,360$. In Python, it is provided by the `math` module as variable `e` with value `2.718281828459045`. In PostgreSQL, you can obtain it via the SQL function `exp(1)` as value `2.718281828459045`[53].

$\mathbb{N_0}$ the set of the natural numbers *including* 0, i.e., 0, 1, 2, 3, and so on. It holds that $\mathbb{N_0} \subset \mathbb{Z}$ and that $\mathbb{N_0} = \mathbb{N_1} \cup \{0\}$.

$\mathbb{N_1}$ the set of the natural numbers *excluding* 0, i.e., 1, 2, 3, 4, and so on. It holds that $\mathbb{N_1} \subset \mathbb{Z}$ and that $\mathbb{N_1} = \mathbb{N_0} \setminus \{0\}$.

$\mathbb{Q}$ the set of the rational numbers, i.e., the set of all numbers that can be the result of $\frac{a}{b}$ with $a, b \in \mathbb{Z}$ and $b \neq 0$. a is called the numerator and b is called the denominator. It holds that $\mathbb{Z} \subset \mathbb{Q}$ and $\mathbb{Q} \subset \mathbb{R}$.

$\mathbb{R}$ the set of the real numbers.

$\mathbb{Z}$ the set of the integers numbers including positive and negative numbers and 0, i.e., ..., -3, -2, -1, 0, 1, 2, 3, ..., and so on. It holds that $\mathbb{Z} \subset \mathbb{R}$.